

OMNIX-RTE-001

Runtime Treasury Execution Trace
ADR-201 . RFC-ATF-1 through RFC-ATF-6

SCENARIO

Autonomous Treasury Approval - Cross-border Liquidity Release

USD 50,000,000 . SWIFT MT202 / XRPL RLUSD

Package: OMNIX-RTE-001-B6B54EEB181F4B05

DUAL-PATH RESULT SUMMARY

PATH DANGEROUS		PATH ADMISSIBLE	
Result	HALT -> OSG REJECTED	Result	ADMITTED -> OSG APPROVED
Budget	42% (drift 758%)	Budget	88% (recertified)
CES	42.0 - CRITICAL	CES	95.5 - NOMINAL
MAS	0.24 - HALT	MAS	0.94 - ALIGNED
Settlement	BLOCKED	Settlement	RELEASED USD 50,000,000

Generated	2026-05-28 08:45:59 UTC
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
Invariants	30 demonstrated
Standard	RFC-ATF-1 through RFC-ATF-6 . ADR-201 . FIPS 204
PQC Public Key SHA-256	AAD3F913EB7C9975DBD4A3C015F32E71202EC294B4B05D63 . . .

§2 - Scenario Overview

Treasury Request

Request ID	REQ-22117C757430
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
Counterparty	-
Counterparty BIC	-
Account (IBAN)	-
Purpose	-
FX Rate	0 EUR/USD
FX Band	-
Risk Level	-
Regulatory Epoch	-

Source State Integrity

source_state_hash41c195cca1cafa51c1f0fb308cc7cd35301102de70ce63a145b70fec9fd6bfe3

Policy Constraints (Governing Mandate)

Max Single-Tx Amount	USD 50,000,000 (requested: USD 50M)
Max FX Deviation	±0%
Counterparty Whitelist	Required - EUROBANK-COUNTERPARTY-001 on list
Dual Approval	Required for amounts > USD 5M
Settlement Window	-

Compliance Tags

Governance Session Parameters

Agent ID	TREASURY-AGENT-OMNIX-001
Agent Role	AUTONOMOUS_TREASURY_EXECUTOR
ATF Layer	L1 (Identity) + L2 (Delegation) + L3 (Temporal) + L4 (Continuity) + BEV L6
Session Type	DUAL-PATH - Dangerous + Admissible
PQC	Dilithium-3 (ML-DSA-65, FIPS 204) . Ephemeral keypair per package
Standards	RFC-ATF-1 -> RFC-ATF-6 . ADR-201

§3 - Governance Architecture

The OMNIX-RTE-001 package exercises the complete ATF + BEV governance stack across two parallel execution paths. Each path traverses an identical 8-step chain, producing a PQC-signed artifact at every transition. The divergence occurs at step 3 (Runtime Evaluation) when the Continuity Execution Score (CES) and MIVP Mandate Alignment Score (MAS) determine whether the session proceeds to settlement or is halted.

8-Step RTE-001 Chain

1_SOURCE_STATE	Source State capture + Temporal Context Snapshot (TCS)
2_AUTHORITY	Delegation Receipt (DR) + Mandate Binding Record (MBR)
3_RUNTIME	Continuity Execution Score (CES) + MIVP Mandate Alignment Score (MAS)
4_COUNTERFACTUAL	Counterfactual Fork Records (CFRs) + Counterfactual Attestation Token (CAT)
5_VERDICT	Temporal Admissibility Record (TAR) - binding verdict
6_GATE	OSG Settlement Gate - ValidationReceipt (VR) + PoGC (admissible only)
7_EXECUTION	Refusal Receipt (dangerous) / BAR + CTCHC + Outcome (admissible)
8_POST_EXECUTION	Replay Proof + Post-execution TCS - terminal forensic seal

Artifact Inventory

Artifact	Name	Hash / Sig Profile	Invariant
DR	Delegation Receipt	SHA-256 + compact + sign(content_hash)	ATF-INV-001
MBR	Mandate Binding Record	SHA3-256 + default + compact sig	MIVP-INV-001
MAS	Mandate Alignment Score	SHA3-256 + default + compact sig	MIVP-INV-003
RCR	Runtime Continuity Record	SHA3-256 + default + compact sig	ATF-INV-003
CFRx5	Counterfactual Fork Records	SHA3-256 per CFR + fragility	CGE-INV-002
CAT	Counterfactual Attestation Token	SHA3-256 + cfr_root_hash + compact sig	CGE-INV-007
TAR	Temporal Admissibility Record	SHA-256 + compact + sign(content_hash)	ATF-INV-004
OSG-VR	OSG Validation Receipt	SHA3-256 + default + compact sig	OSG-INV-001
BAR	Behavioral Anchor Record	SHA3-256 6-field canonical + default sig	BEV-INV-002
CTCHC	Cross-Turn Coherence Hash Chain	SHA3-256 per link + 7-field seal hash	BEV-INV-013
PoGC	Proof of Governance Certificate	SHA3-256 + compact sig	PoGR-INV-003
MBR Seal	MBR Session Seal	SHA3-256 + compact sig + tier	MIVP-INV-007
TCS	Temporal Context Snapshot	SHA3-256 + default + compact sig	TGB-INV-001
Replay	Post-execution Replay Proof	SHA3-256 + default + compact sig	RTE-INV-007

§4 - Dual-Path Comparison		
Dimens	PATH DANGEROUS	PATH ADMISSIBLE
Authori	42% (drift ?58%)	88% (recertified)
DR Expi	~22 min (near-expiry)	4h (fresh)
CES Sc	42.05	95.50
CES Ba	CRITICAL	NOMINAL
MIVP M	0.24 (HALT < 0.30)	0.94 (ALIGNED > 0.65)
MIVP Ve	HALT	ALIGNED
TAR Sta	ADMITTED	-
OSG Ve	REJECTED	APPROVED
BAR Sta	VALID	VALID
CTCHC	True	True
PoGC Is	No (HALT path)	POGC-68BE5F67982E417D
Mandati	UNCERTIFIED	MANDATE-BOUND
Settlem	BLOCKED	RELEASED - T+2 clearing
Replay	HALTED	CLOSED

§5 - Authority Layer: DR + MBR (Both Paths)

PATH DANGEROUS - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-96579FDED8E74F89
Budget Granted	42.0%
Budget Delegator	100.0%
MAR Reduction	?58.0%
Issued At	-
Expires At	2026-05-28 09:07:42 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	ee34cc16c7f67cddf452f3cc03fe22817ad8f30e5c860211172efd7707a3d44a
pqc_signature	mtux069a3eTZJtw+AS1RCuWa11Uw9ED5+6mP3bRBksAdPdfPr4I53B00x2aYGu6ENHxqWuCm

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-EF1A089560AA4F24
Session ID	SESSION-9954D0EDF7624EEC
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-28 08:45:43 UTC
mbr_content_hash	e31fb5414201154e19c2fc83f4fdd9094912acddd516bc73388443ce6b5532ef

PATH ADMISSIBLE - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-FD3B4FED63FF436A
Budget Granted	88.0%
Budget Delegator	100.0%
MAR Reduction	?12.0%
Issued At	-
Expires At	2026-05-28 12:45:50 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	50c734b1f91baf18b0115e754c2c6470065f07bec5da3b2857843f209fe96cc
pqc_signature	ZiC4ZSYEmvbP/NEBDjO/Jz3PjRDi7oX0vsSfnk96I8RE/tgdVgv1CyKu4kLDt09lFo+hz9e

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-D21BA8EED3E84A97
Session ID	SESSION-8C7E9772F5FD4F4E
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-28 08:45:51 UTC
mbr_content_hash	74e61b8df8cc2bc6412fd9b9ecab10381272ca8d26878f7fe63985295c17fa8e

§6 - Runtime Evaluation: RCR / CES + MAS (Both Paths)

PATH DANGEROUS - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-0796BCAE25BC403F
CES Score	42.05
CES Band	CRITICAL
T (temporal)	9.2% (weight 0.30) -> 2.75
B (budget)	42.0% (weight 0.30) -> 12.60
D (context)	61.5% (weight 0.20) -> 12.30
I (integrity)	72.0% (weight 0.20) -> 14.40
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 42.05$
Issued At	2026-05-28 08:45:43 UTC
rcr_hash	f9e3aa9b283eaf73b73f41cb823893a85ec3940424017d2abadc768f1605e4

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-89ADE93092974DC7
Turn Index	0
Score	0.2400
Verdict	HALT
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-28 08:45:44 UTC
mas_content_hash	47d789f50788c01626ca613d78d0b048108c1d926b9520637efa50ea79573dfc

PATH ADMISSIBLE - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-6B51D1E12B1E4AC4
CES Score	95.50
CES Band	NOMINAL
T (temporal)	100.0% (weight 0.30) -> 30.00
B (budget)	88.0% (weight 0.30) -> 26.40
D (context)	97.0% (weight 0.20) -> 19.40
I (integrity)	98.5% (weight 0.20) -> 19.70
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 95.50$
Issued At	2026-05-28 08:45:51 UTC
rcr_hash	15e98160d80e7126e56da294f4721474ff346719a84376746083dc4add228a30

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-F8CE12D595BA4325
Turn Index	0
Score	0.9400
Verdict	ALIGNED
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-28 08:45:52 UTC
mas_content_hash	67c36cdce399cc253c5c879e116e53ddb4c9a98c4244459ff879d6fe99204025

\$7 - Counterfactual Analysis: CGE CFRs + CAT (Both Paths)

PATH DANGEROUS - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-70FBFBA5520D49B4
CFR Count	5
cfr_root_hash	b854cdd67eaddac3e437cf25a663c9cdfe616bdc2e785eedd065b702...
Issued At	2026-05-28 08:45:44 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-0B564914FE014A0A	-	0.94	YES	ADMITTED
CFR-FD3AE538F39545A1	-	0.08	no	ADMITTED
CFR-D5B6466CE0A34082	-	0.71	no	ADMITTED
CFR-A3398CD867404620	-	0.85	no	ADMITTED
CFR-1DED4EFD39834504	-	0.88	no	ADMITTED

PATH ADMISSIBLE - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-0D8A938BB0404DCA
CFR Count	5
cfr_root_hash	abc48c6edd41937183e5eb443f7fa913c23a23d16ac33eab805e3f79...
Issued At	2026-05-28 08:45:52 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-137F2D5A768C4D5C	-	0.09	YES	ADMITTED
CFR-CFDD8B23F3A7A47D6	-	0.14	no	ADMITTED
CFR-108ACF9D790341FB	-	0.18	no	ADMITTED
CFR-313EBF74A8AC4108	-	0.11	no	ADMITTED
CFR-557591CF34F44CB8	-	0.13	no	ADMITTED

§8 - DANGEROUS Path: HALT Evidence Chain

Refusal Receipt . RTE-INV-002

Receipt ID	REFUSAL-102B830EF07F4BF5
Type	HARD_REFUSAL
Settlement Status	BLOCKED - OSG will independently reject
Mandate Cert	UNCERTIFIED
Rejection Reasons	5
Issued At	2026-05-28 08:45:45 UTC

Rejection Reasons:

- ? CES=42.05 - CRITICAL band (HALT threshold for CRITICAL-risk transaction > USD 10M)
- ? MIVP MAS=0.24 - below halt_threshold=0.3 (MIVP-INV-003)
- ? MIVP proxy_guard_violation: Speed-over-verification optimisation detected
- ? TAR: Authority budget 42% insufficient for CRITICAL-risk USD 50M transaction
- ? CCS: cumulative drift CRITICAL - 3 consecutive drift signals before Turn 1

OSG Validation Receipt (REJECTED) . OSG-INV-001

VR ID	VR-EBAC1D60C96348F6
Verdict	REJECTED
Fail Closed	True
Issued At	2026-05-28 08:45:45 UTC
vr_content_hash	a71f42f52df0850984258e46343f83c1bfb8db3c62b1a9933766c7c62d5a83bf

MBR Session Seal (UNCERTIFIED) . MIVP-INV-007

Seal ID	MBRSEAL-1D126F9D7BE84D1E
Certification Tier	UNCERTIFIED
Issued At	2026-05-28 08:45:45 UTC
seal_content_hash	74ce680e0bfda66fcdb19c595e7e820e9fd5cd7e499d1cb93c7327ee58f46e78

Behavioral Anchor Record (BAR) . BEV-INV-001

BAR ID	BAR-BD0A000D80444F3D
Status	VALID
Halt Reason	-
Turn Index	0
content_hash	59338e6f1bee623815163f30936e436ade640d4c90069fcfa0f66eb8ad19a882
output_hash	437fbec2834d845c077604bc0e483f50eb098f969a7daad0fbf6b7eff72d66bf

CTCHC - Sealed in HALTED State . BEV-INV-013

Chain ID	CTCHC-634BAF9E756D4FF2
Turn Count	1
Is Sealed	True
Sealed At	2026-05-28 08:45:49 UTC
seal_hash	198075ea5d5896a2ca075606987e7074ca723cf59fe5679411d610f2380ec245

Structural Invariants

execution_occurred	False
settlement_released	False

§9 - ADMISSIBLE Path: Settlement Evidence Chain

TAR . Temporal Admissibility Record - ADMITTED . ATF-INV-004

TAR ID	-
Admission Status	ADMITTED
Execution ns	0
Issued At	-
content_hash	-

TAR Binding Record

Binding ID	BIND-B2337CA76E11
Status	-

Proof of Governance Certificate (PoGC) . PoGR-INV-003

PoGC ID	POGC-68BE5F67982E417D
Mandate Certification	MANDATE-BOUND
TAR Status	ADMITTED
BAR Status	VALID
CTCHC Turn Count	1
Standard	RFC-ATF-6 (ADR-181/182/183/186) . ADR-194 (MIVP)
Regulatory Tags	EU AI Act Art. 9, MiCA Title VI, DORA Art. 11
Issued At	2026-05-28 08:45:59 UTC
content_hash	e54c8e24ee137d45663366242ba3e0037720b0c5e997c465ddf79f4b46d31e3e

MBR Session Seal (MANDATE-BOUND) . MIVP-INV-008

Seal ID	MBRSEAL-983350D972E649FC
Certification Tier	MANDATE-BOUND
Issued At	2026-05-28 08:45:59 UTC
seal_content_hash	946c57128914e6a9c9b1ec3df534b505f3c6e2b0f2469a1d3a6ee128560f707c

OSG Validation Receipt (APPROVED) . OSG-INV-001

VR ID	VR-15A4A312EC6B4EC1
Verdict	APPROVED
Fail Closed	True
Issued At	2026-05-28 08:45:59 UTC
vr_content_hash	5e7be1b22abfd7cc7c901d0319d7ef71aae4019f253fa6be7554c011a4c3a3de

Settlement Reference

Amount (USD)	USD 50,000,000
SWIFT MT202 Ref	MT202-79276C89D6CB
XRPL TxID	XRPL-599E969416944B7D975D6D30
Clearing	-
FX Rate	-
Counterparty	-
IBAN	-

Outcome Receipt . RTE-INV-004

Outcome ID	-
Type	ADMISSION_OUTCOME
Mandate Cert	MANDATE-BOUND
Issued At	2026-05-28 08:45:59 UTC
content_hash	fa4bfff3b893e9b50880cfb3b6066b876a8a3186e4cf977e5df6cc3231d50546b

§10 - Post-Execution: TCS + Replay Proofs (Both Paths)

PATH DANGEROUS - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-0CE5A53A97E14447
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	3eef26c05e19d9c2af01fc656a57aaf07ba0aa4f8608d70768d3ea9311dac275

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-8EAA63E3179D42CB
Terminal Status	HALTED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-634BAF9E756D4FF2
CTCHC Seal Hash	198075ea5d5896a2ca075606987e7074ca723cf5...
Turn Count	1
Issued At	2026-05-28 08:45:50 UTC
proof_content_hash	9f00461733f8792a04aaa22a91f700d75cb363beb849bf6bb14ae85003b692c3

PATH ADMISSIBLE - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-571C2AD06AAD4BDA
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	9168a8ba12231abf1657b1727d68abd19d2da2b6b93b7b81694b50988cfd271c

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-74577187752941B6
Terminal Status	CLOSED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-E40E13ACA09144A8
CTCHC Seal Hash	2cf94518b531accecea2e4d2faac43f5a41efc34...
Turn Count	1
Issued At	2026-05-28 08:45:59 UTC
proof_content_hash	8d778f63903323867ba89b9c829b1fc7bd362a6e1384717f451ec4c1a4f57dce

§11 - Offline Verification: CLI Report

The following verification was produced by the standalone CLI verifier (scripts/verify_treasury_execution_trace.py) with no runtime dependencies. Verification requires only the public key embedded in the package.

Verification Commands

```
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json --verify-authority
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json --verify-continuity
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json --verify-halt
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json --verify-settlement
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_084559.json --json
```

Verification Result - 111 / 111 Checks

PASS	[PKG-TYPE] package_type=OMNIX-RTE-001
PASS	[PKG-PATHS] Both dangerous and admissible paths present
PASS	[PKG-PQC] Embedded public_key_b64 present
PASS	[PKG-INV] 30 invariants demonstrated (>=20 required)
PASS	[DR-DAN-HASH] DANGEROUS DR content_hash integrity
PASS	[DR-DAN-SIG] DANGEROUS DR PQC signature (ML-DSA-65)
PASS	[DR-DAN-MAR] DANGEROUS DR MAR: 42.0 <= 100.0 (ATF-INV-001)
PASS	[MBR-DAN-SIG] DANGEROUS MBR PQC signature (MIVP-INV-001)
PASS	[RCR-DAN-CES] CES formula consistent: 42.05
PASS	[RCR-DAN-BAND] CES band CRITICAL (dangerous path)
PASS	[MAS-DAN-SIG] DANGEROUS MAS PQC signature (MIVP-INV-003)
PASS	[CAT-DAN-ROOT] CAT cfr_root_hash covers all CFR IDs
PASS	[CAT-DAN-COUNT] CAT cfr_count >= 3 (got 5)
PASS	[CAT-DAN-SIG] CAT PQC signature (CGE-INV-007)
PASS	[REF-SIG] Refusal receipt PQC signature
PASS	[REF-TYPE] type=HARD_REFUSAL
PASS	[REF-SETTLE] settlement_status=BLOCKED
PASS	[BAR-DAN-HASH] DANGEROUS BAR content_hash (BEV-INV-002)
PASS	[BAR-DAN-SIG] DANGEROUS BAR PQC signature (BEV-INV-004)
PASS	[CHC-DAN-CHAIN] CTCHC link chain continuity (BEV-INV-011)
PASS	[CHC-DAN-SEAL] CTCHC seal_hash covers all links (BEV-INV-013)
PASS	[CHC-DAN-SIG] CTCHC seal PQC signature (BEV-INV-014)
PASS	[OSG-DAN-VERDICT] OSG verdict=REJECTED
PASS	[HALT-EXEC] execution_occurred=False (RTE-INV-002)
PASS	[HALT-SETTLE] settlement_released=False
PASS	[POGC-SIG] PoGC PQC signature (PoGR-INV-003)
PASS	[POGC-BOUND] mandate_certification=MANDATE-BOUND
PASS	[SEAL-ADM-TIER] MBR Seal tier=MANDATE-BOUND (MIVP-INV-008)
PASS	[OSG-ADM-VERDICT] OSG verdict=APPROVED
PASS	[BAR-ADM-SIG] ADMISSIBLE BAR PQC signature (BEV-INV-004)
PASS	[CHC-ADM-SIG] ADMISSIBLE CTCHC seal PQC signature (BEV-INV-014)
PASS	[TAR-ADM-SIG] ADMISSIBLE TAR PQC signature (ML-DSA-65)
PASS	[OUT-SIG] Outcome receipt PQC signature
PASS	[SETTLE-AMOUNT] Settlement amount = USD 50,000,000
PASS	[RPL-DAN-STATUS] Replay proof terminal_status=HALTED
PASS	[RPL-ADM-STATUS] Replay proof terminal_status=CLOSED
PASS	[RPL-DAN-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)
PASS	[RPL-ADM-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)

VERDICT: 111 / 111 VERIFICATIONS PASS - package integrity confirmed

§12 - Invariants Demonstrated

This package demonstrates 30 governance invariants across RFC-ATF-1 through RFC-ATF-6 and ADR-201. Each invariant is verified by the standalone CLI verifier without platform access.

Invariant
ATF-INV-001: Authority never expands through delegation (MAR enforced at DR issuance)
ATF-INV-002: Every DR PQC-signed by delegator (Dilithium-3)
ATF-INV-005: Receipts immutable once issued (content_hash)
RGC-INV-001: Every RCR anchored to a valid TAR
RGC-INV-002: CES computed from real-time component values
BEV-INV-001: Every governed turn produces a BAR before output is delivered
BEV-INV-010: CTCHC initialized before first BAR
BEV-INV-011: Each CTCHC link = H(prev turn_hash governing_receipt_id)
BEV-INV-013: Seal hash covers complete chain (first->last link)
BEV-INV-014: CTCHC seal is PQC-signed (ML-DSA-65) before OEP export
MIVP-INV-001: MBR issued before Turn 1 (mandate frozen at session open)
MIVP-INV-003: MAS computed per turn linked to CTCHC link hash
MIVP-INV-007: MBR Seal issued at session close
MIVP-INV-008: MANDATE-BOUND tag on PoGC (admissible path, zero violations+warnings)
CGE-INV-001: CFRs computed at evaluation time (not retroactively)
CGE-INV-002: CAT root hash covers all CFR IDs
CGE-INV-003: fragility_score ? [0.0, 1.0]
CGE-INV-007: CAT PQC-signed before OEP export
TGB-INV-001: TCS embedded in source_state with nanosecond-precision regulatory context
PoGR-INV-001: PoGC issued only from sealed CTCHC session
PoGR-INV-002: PoGC append-only - content_hash immutable after issuance
PoGR-INV-003: PoGC verifiable offline (zero runtime, zero auth)
RTE-INV-001: Package contains both dangerous and admissible paths
RTE-INV-002: Dangerous path terminates in HALT (steps 3+5)
RTE-INV-003: OSG rejects dangerous path independently (fail-closed)
RTE-INV-004: Admissible path PoGC carries MANDATE-BOUND certification
RTE-INV-005: Both paths contain CGE CAT with 5 CFRs
RTE-INV-006: Both paths contain TCS with regulatory_context
RTE-INV-007: Package verifiable offline - zero OMNIX runtime required
RTE-INV-008: CLI verifier exits 0 on PASS, non-zero on FAIL

§13 - Canonicalization Profile & Technical Notes

Canonicalization Profile - ADR-200 §4 + ADR-201

Artifact	Hash	JSON Sep	Sig Payload	Invariant
DR / TAR	SHA-256	compact (',',':')	sign(content_hash.encode())	ATF stack
RCR	SHA3-256	default	compact sig {rcr_id, rcr_hash}	ATF-INV-003
MBR / MAS	SHA3-256	default (excl key)	compact sig {id, hash, issued_at}	MIVP-INV-001/003
MBR Seal	SHA3-256	default	compact sig {seal_id, hash, issued_at}	MIVP-INV-007
CAT	SHA3-256	default	compact sig {cat_id, hash, issued_at}	CGE-INV-007
TCS / Replay	SHA3-256	default	compact sig {id, hash, issued_at}	TGB-INV-001
OSG VR	SHA3-256	default	compact sig {vr_id, hash, issued_at}	OSG-INV-001
Refusal / Outcome	SHA3-256	default	compact sig {receipt_id, hash, issued_at}	RTE-INV
BAR content_hash	SHA3-256	default (6 fields)	default sig {bar_id, hash, governing_receipt_id, created_at}	BEV-INV-002
CTCHC link hash	SHA3-256	default	SHA3({prev, turn, receipt})	BEV-INV-011
CTCHC seal_hash	SHA3-256	7-field payload	default sig {chain_id, seal_hash, session_id}	BEV-INV-013
PoGC	SHA3-256	default (incl alg)	compact sig {pogc_id, hash, issued_at}	PoGR-INV-003

Verifier Scope Limits

OUT OF SCOPE	Governance policy value validation (FX bands, counterparty lists, mandate amounts)
OUT OF SCOPE	External market data in source_state
OUT OF SCOPE	OMNIX runtime, database, or network access
IN SCOPE	Embedded public key produces valid PQC signatures for ALL artefacts (30 sig checks)
IN SCOPE	Hash integrity confirmed for all 14 artifact types
IN SCOPE	Dangerous path HALTED: execution_occurred=False, settlement_released=False
IN SCOPE	Admissible path CLOSED: execution_occurred=True, settlement_released=True
IN SCOPE	CTCHC chain continuity turn-by-turn verification
IN SCOPE	CAT cfr_root_hash covers all 5 CFR IDs
IN SCOPE	CES formula consistency: Tx0.30 + Bx0.30 + Dx0.20 + lx0.20

§14 - Verification Summary & PQC Public Key

Package Summary

Package ID	OMNIX-RTE-001-B6B54EEB181F4B05
Package Type	OMNIX-RTE-001
Generated At	2026-05-28 08:45:59 UTC
Scenario	Autonomous Treasury Approval
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
PQC Key Size PK	0 bytes (Dilithium-3 standard)
PQC Key Size SK	0 bytes
Total Artifacts	14 types . 2 paths . 30 PQC signatures
Invariants	30 demonstrated
Verifier Checks	111 / 111 PASS
ADR Reference	ADR-201 . RFC-ATF-1 -> RFC-ATF-6
FIPS Standard	FIPS 204 (ML-DSA-65 / Dilithium-3)

PQC Public Key (ML-DSA-65) - Embedded for Offline Verification

The following public key is embedded in the package. It is the sole key required to verify all 30 PQC signatures in this package offline. No other key material, network access, or OMNIX runtime is required.

SHA-256 fingerprint	AAD3F913EB7C9975DBD4A3C015F32E71202EC294B4B05D63FEF2694C78A6F47D
SHA3-256 fingerprint	54CFEAE44B426B327AA496D6C38AB61F7D92A7852BB4342C5D21826846BA1809

Public Key (Base64, first 512 chars):

XEKUZgo98L0lJeMqWllvVc3CvgnyKllrE3H7sdC0x2zy8Lemxtfi23ad0ZeMcIW9TjNJmKbzIVZc6LgDnxgfwuTNnaJGG10YQPSt/VtaUllxPp4+SptE4BevbJYGiuBnie3zMix95JcrLF5FtjdYbud0ovAlhelkwcuXACa0yOE+iw270qYXT71HMDiU/vd3xm0DCmGUu2T8zmNNFjeN5ABRhADx04Z2yl3pcrizKt849gngDk7yVPIqa7JbiMu0geyEOxzNJsnp58peJotJ/6nCSw0DWR4vxXmepXWLFa0S+mHEP5wRqyZZG+jdiferfAjYS6TVc1N5CB0QaungSCL0x0nuw0kgivnJMC4Xbw8nC9Vlqgfy5P9rvUC3DT9Wv60nuMLX2ZiipD63C8MMqEItQAjnv5uuxTptXy8aTNLGsue0B/8TSUBdx+Mcn3BjWIOXaDEFe4lhroCEdsG+iuVEA6+NJpiem9D0jedSCw4ZWqFitUcu eRxxwQ2XxjcNm

Governance Verdict

PATH DANGEROUS: HALT PROVEN	PATH ADMISSIBLE: SETTLEMENT RELEASED
OMNIX-RTE-001 . 111 / 111 VERIFICATIONS PASS . RFC-ATF-1 -> RFC-ATF-6	